CHARAN BS

AI / ML Engineer & Application Developer

Email : charan201204@gmail.com

Phone : +91 91130 10019

Location : Hassan, Karnataka

Portfolio : charan-bs-website.onrender.com

GitHub : github.com/CharanBS18

LinkedIn : linkedin.com/in/charan-bs-b34975391

PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with hands-on experience designing and deploying intelligent applications using Python, LangChain, RAG pipelines, and OpenAI APIs. Demonstrated ability to build, integrate, and ship machine learning-powered tools into real-world, production-grade systems — including a live AI diagnostic application serving real users. Passionate about automation, deep learning, and staying at the forefront of Artificial Intelligence to build solutions that make a real-world impact.

EDUCATION

Bachelor of Engineering (AI & ML)
Rajeev Institute of Technology
Hassan, Karnataka
Relevant Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems, AI/ML, Statistics

TECHNICAL SKILLS

AI / ML & Deep Learning

Machine Learning	Deep Learning	Neural Networks	NLP
RAG Pipelines	LangChain	TensorFlow	PyTorch

Programming & Frameworks

Python	Streamlit	FastAPI	REST APIs
OpenAI APIs	Hugging Face	scikit-learn	Pandas / NumPy

Data Engineering & MLOps

Data Collection	Data Cleaning	Data Preprocessing	Feature Engineering
Model Evaluation	Vector Databases	Git / GitHub	Render / Vercel

PROJECTS

Plant Health RAG — AI Diagnostic Application

Tech / Tools: Python, Streamlit, LangChain, RAG, OpenAI API, Vector Database, Deployed on Render

- Designed and deployed a full AI-powered plant disease diagnostic app using a Retrieval-Augmented Generation (RAG) pipeline.
- Implemented complete data pipeline: document ingestion, text chunking, vector embedding, semantic retrieval, and LLM answer generation.
- Integrated OpenAI API for natural language responses grounded in domain-specific agricultural knowledge.
- Evaluated model outputs for accuracy and relevance; iterated on chunking strategy and retrieval parameters to improve results.
- Live, production deployment: planthealthrag-fsggg4j6desaboqkfomqeb.streamlit.app

Lumina Volt — EV Charging Station Finder

Tech / Tools: Python, REST APIs, Geolocation API, JavaScript, WebGL

• Built a geolocation-powered EV station finder integrating REST APIs for real-time station data retrieval.
• Implemented dynamic route pathing and distance calculation between user location and nearest available stations.
• Collaborated across the full stack to deploy a premium interactive frontend with 3D route visualisation.
• Live: charanbs18.github.io/EV-project-PPT

AI Chatbot & Automation Experiments

Tech / Tools: Python, OpenAI API, LangChain, Streamlit, REST APIs

• Built conversational chatbot prototypes using OpenAI API and LangChain with persistent memory and multi-turn context management.
• Explored automation pipelines including web scraping, data preprocessing, and AI-triggered scheduled workflows.
• Applied prompt engineering techniques to optimise response quality, reduce hallucination, and control output format.

ALIGNMENT WITH ROLE REQUIREMENTS

• AI-Powered Application Development	Built & deployed a production RAG chatbot/diagnostic app; developed chatbot and automation tools using Python and OpenAI APIs.
• ML / DL Model Building & Optimisation	Hands-on with scikit-learn, TensorFlow, and PyTorch for model training, tuning, and performance optimisation.
• Large Dataset Handling	Experienced in data collection, cleaning, preprocessing, and feature engineering using Pandas and NumPy.
• Python Frameworks — TensorFlow, PyTorch	Proficient in Python; implemented models using TensorFlow, PyTorch, LangChain, and Hugging Face Transformers.
• OpenAI API & REST API Integration	Integrated OpenAI API in a live production system; built and consumed REST APIs across multiple deployed projects.
• Deploying AI into Real-World Systems	Deployed AI apps on Render and Streamlit Cloud; experienced working across frontend/backend layers.
• Model Evaluation & Performance Testing	Applied accuracy, F1, precision/recall metrics; iteratively optimised RAG retrieval parameters and prompt strategies.
• Staying Updated with AI Trends	Actively follows LLMs, RAG, vector databases, and multimodal AI; applies emerging techniques directly to live projects.

SOFT SKILLS

Analytical Thinking	Problem Solving	Self-Directed Learning	Attention to Detail
Collaboration	Communication	Adaptability	Initiative

LANGUAGES

English — Professional **Kannada** — Native **Hindi** — Conversational

DECLARATION

I hereby declare that all the information furnished above is true and correct to the best of my knowledge and belief.

Place: Hassan, Karnataka

Charan BS